

Hyun Jeong (鄭 玄)

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Personal Statement

I am a Ph.D. student studying gravity both from the phenomenological and theoretical perspectives. My research began with inflationary cosmology, with a particular focus on its observational consistency, while deepening my understanding of the underlying theoretical challenges. Recently, I have been exploring scattering amplitudes and their applications to black hole physics, and I look forward to advancing this direction through discussions with you all.

Education

Apr 2024 – present	Ph.D. in Physics (in progress), The University of Tokyo Advisor: Prof. Jun'ichi Yokoyama
Apr 2022 – Mar 2024	M.Sc. in Physics, The University of Tokyo Advisor: Prof. Jun'ichi Yokoyama Thesis: “Cosmological Evolution after R^2 inflation”
Apr 2018 – Mar 2022	B.Sc. in Physics, The University of Tokyo

Research Experience

Feb 2026 – Mar 2026	Visiting Researcher, National Taiwan University Host: Prof. Yu-Tin Huang
Jan 2025 – Mar 2025	Visiting Researcher, YITP, Kyoto University Host: Dr. Katsuki Aoki

Grants and Fellowship

Apr 2024 – Mar 2027	JSPS Research Fellowship for Young Scientists (DC1) , Japan Society for the Promotion of Science (JSPS), Grant No. 24KJ0902.
Apr 2022 – Mar 2027	Forefront Physics and Mathematics Program to Drive Transformation (FoPM) , The University of Tokyo.

Languages

Japanese (native). English and Korean (fluent). German (basic).

Publications

- K. Aoki, F.-Y. Cheng, A. Cristofoli, Y.-t. Huang and **H. Jeong**, An Effective S-matrix Approach to Low-Frequency Waveforms from Black Hole Mergers, [arXiv: 2609.10329].
- D. Akpinar, K. Aoki, A. Cristofoli, **H. Jeong** and K. Yoshimura, *Black Hole Thermodynamics Meets On-Shell Amplitudes: Local Detailed Balance and Thermal Spectrum from Spin Universality and Unitarity*, [arXiv: 2606.13599].
- K. Aoki, A. Cristofoli, **H. Jeong**, M. Sergola and K. Yoshimura, *Quantum Effects for Black Holes with On-Shell Amplitudes*, *JHEP* **12** (2025) 163.
- H. Jeong**, K. Kamada, A. A. Starobinsky and J. Yokoyama, *Reheating process in the R^2 inflationary model with the baryogenesis scenario*, *JCAP* **11** (2023) 023.

Presentations (International Conferences)

1. **H. Jeong**, “Gravitational Memory in Black Hole Mergers from Scattering Amplitudes”, Theoretical Developments in Gravitational Wave Physics, Kyoto University, Jun 15-19, 2026 (oral).
2. **H. Jeong**, “Quantum Effects for Black Holes with On-Shell Amplitudes”, Nagoya workshop on General Relativity, Nagoya University, Jan 26-28, 2026 (oral).
3. **H. Jeong**, “Quantum Effects for Black Holes with On-Shell Amplitudes”, The 34th Workshop on General Relativity and Gravitation in Japan (JGRG 34), Kyoto University, Jan 19-23, 2026 (oral).
4. **H. Jeong**, “The role of Higgs field after R^2 inflation”, COSMO’24, Kyoto University, Oct 21-25, 2024 (poster).
5. **H. Jeong**, “Are primordial black holes produced after R^2 inflation?”, The 32nd Workshop on General Relativity and Gravitation in Japan (JGRG 32), Nagoya University, Nov 27-Dec 1, 2023 (oral).
6. **H. Jeong**, “ R^2 inflation with the baryogenesis scenario”, FoPM International Symposium, The University of Tokyo, Feb 6-8, 2023 (poster).

Presentations (Domestic Conferences)

1. **H. Jeong**, “Gravitational Memory in Black Hole Mergers from Scattering Amplitudes”, 81th Annual Meeting of The Physical Society of Japan, The University of Tokyo, Sep 14-17, 2026 (oral).
2. **H. Jeong**, “Effects of Hawking Radiation on Black Hole Scattering”, 80th Annual Meeting of The Physical Society of Japan, Hiroshima University, Sep 16-19, 2025 (oral).
3. **H. Jeong**, “The role of Higgs field after R^2 inflation”, Joint workshop on General Relativity and Cosmology 2025, Nishiyatsu-Onsen, Mar 10-11, 2025 (oral).
4. **H. Jeong**, “The role of Higgs field after R^2 inflation”, 79th Annual Meeting of The Physical Society of Japan, Hokkaido University, Sep 16-19, 2024 (oral).
5. **H. Jeong**, “Are primordial black holes produced after R^2 inflation?”, 2024 Spring Meeting of The Physical Society of Japan, online, Mar 18-21, 2024 (oral).
6. **H. Jeong**, “Toward a theoretically and phenomenologically consistent model”, Joint workshop on General Relativity and Cosmology 2024, Sunshine Shirako, Mar 6-8, 2024 (oral).
7. **H. Jeong**, “Reheating process in the R^2 inflationary model with the baryogenesis scenario”, 78th Annual Meeting of The Physical Society of Japan, Tohoku University, Sep 16-19, 2023 (oral).
8. **H. Jeong**, “ R^2 inflation with the baryogenesis scenario”, Joint workshop on General Relativity and Cosmology 2023, Okutama Town Social Welfare Council, Mar 13-14, 2023 (oral).
9. **H. Jeong**, “ R^2 inflation with the baryogenesis scenario”, The 35th Rironkon Symposium, Corasse Fukushima, Dec 21-23, 2022 (poster).

Seminars

1. **H. Jeong**, “An Effective S-matrix Approach to Black Hole Mergers”, Niels Bohr Institute, University of Copenhagen, Oct 5, 2026.
2. **H. Jeong**, “An Effective S-matrix Approach to Black Hole Mergers”, Higgs Centre for Theoretical Physics, The University of Edinburgh, Oct 2, 2026.
3. **H. Jeong**, “An Effective S-matrix Approach to Black Hole Mergers”, CTP Seminar, Queen Mary University of London, Sep 24, 2026.
4. **H. Jeong**, “Quantum Effects for Black Holes with On-Shell Amplitudes” (blackboard presentation-style), NTU Strings Seminar, National Taiwan University, Mar 18, 2026.
5. **H. Jeong**, “Quantum Effects for Black Holes with On-Shell Amplitudes” (slide presentation-style), NTU Strings

Awards

1. **H. Jeong**, Outstanding Presentation Award Gold Prize (out of 112), The 34th Workshop on General Relativity and Gravitation in Japan (JGRG 34), Kyoto University, Jan 19-23, 2026.
2. **H. Jeong**, Outstanding Presentation Award Gold Prize (out of 14), Joint workshop on General Relativity and Cosmology 2023, Okutama Town Social Welfare Council, Mar 13-14, 2023.
6. **H. Jeong**, Oral Award (2nd place out of 39), Gravity & Cosmology Section, “Starobinsky Model vs. Higgs Inflation Model”, The 52nd Astronomy & Astrophysics Summer School for Young Researchers, Aug 23-26, 2022.
7. Special Prize: Japanese Institute of Certified Public Accountants Award, Mathematics Koshien 2017 – As a Team Leader of Toin Gakuen Secondary Education School (Pre-university).

Outreach and Writing

1. [Student essay, “Black Holes as an Energy Source”, FoPM graduate course “Scientific Writing, Publication, and Communication” \(FY2022\).](#)
2. [“Singularity Theorems and Future Prospects”, Physics Lab. \(Dept. of Physics\) Advent Calendar 2020, Day 22 \(in Japanese\).](#)
3. [“Convergence Factors from the Perspective of Time Ordering”, Physics Lab. \(Dept. of Physics\) Advent Calendar 2020, Day 19 \(in Japanese\).](#)

Referees

Prof. Jun'ichi Yokoyama | The University of Tokyo | junichi.yokoyama@ipmu.jp

Prof. Kohei Kamada | Hangzhou Institute for Advanced Study, UCAS | kohei.kamada@ucas.ac.cn

Dr. Katsuki Aoki | Kyoto University | katsuki.aoki@yukawa.kyoto-u.ac.jp